Asst. Professor Ms. Harshita Pandey

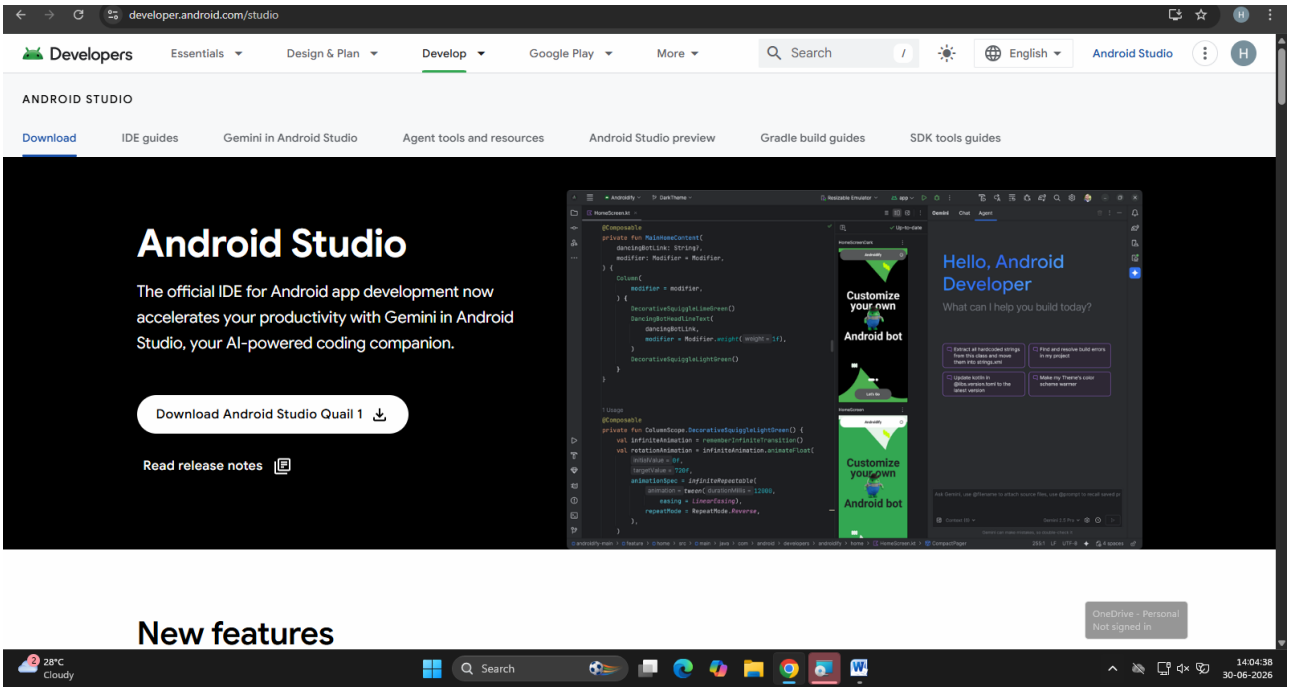
UBIQUITOUS COMPUTING

Practical No. 1

AIM: Design and Develop **Location-Based Messaging App**

STEPS:

Step1:



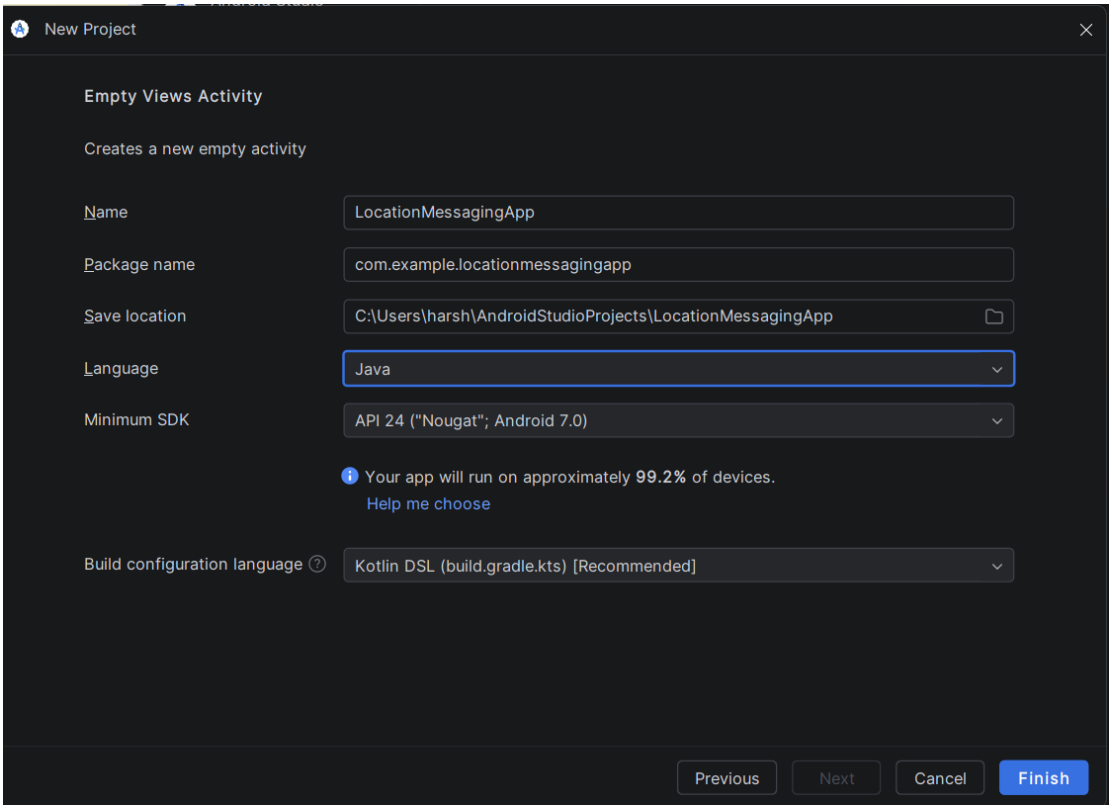
Step2:

Projects > new project > phone and tablet > Empty Views Activity

Do NOT choose "Empty Activity", which is not suitable for your Java practical.

Step3:

Field	Value
Name	`LocationMessagingApp`
Package Name	Leave the default
Save Location	Leave the default
Language	**Java**
Build Configuration Language	**Kotlin DSL** (default)
Minimum SDK	**API 24: Android 7.0 (Nougat)**

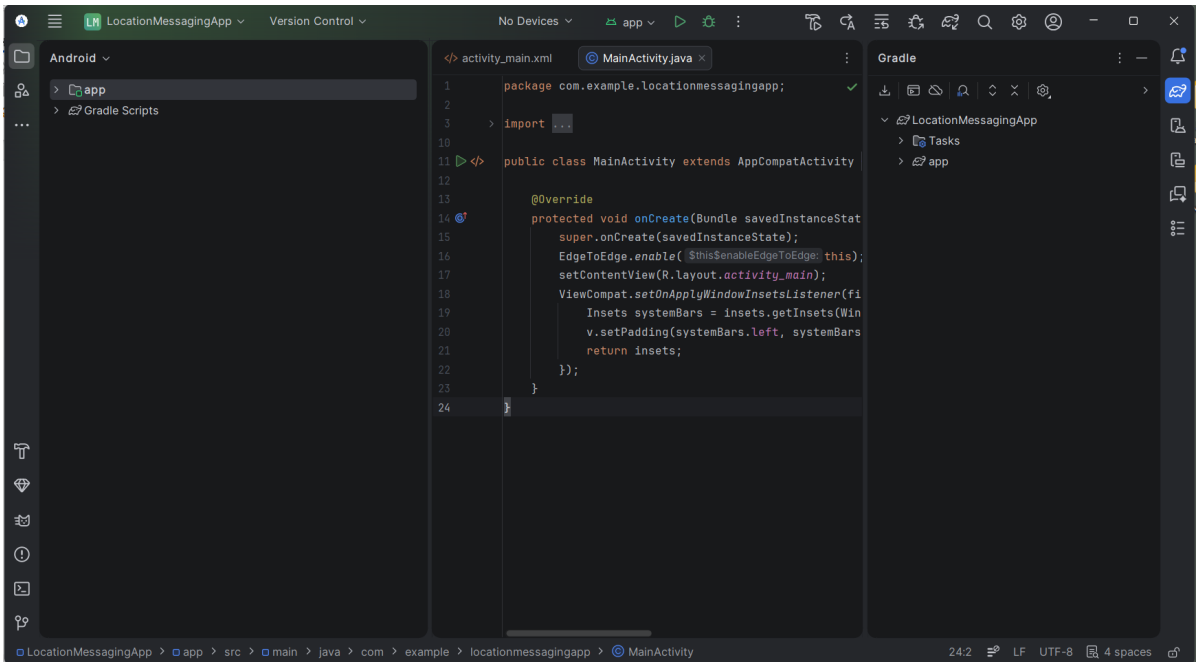


Step4:

Setup SDK

Step5:

Your project is now **created successfully**.



I can see:

- Gradle sync completed.
- JDK issue is resolved.
- `MainActivity.java` is open.

- `activity_main.xml` is open.
- No errors.

If you cannot see the folder, then open it manually:

Main activity.java

```
kotlin+java
 └── com.example.locationmessagingapp
      ├── MainActivity.kt
```

activity_main.xml

```
app
 └── res
      └── layout
           └── activity_main.xml
```

The only thing missing is **No Devices**, which means you haven't created an Android emulator yet.

Step6:

If you want to **run your Android Studio project on your physical Android phone via USB** (instead of using the Android Emulator/AVD), follow these steps.

For your **phone**, follow these steps:

Step 1: Enable Developer Options

1. Open **Settings**.
2. Tap **About Phone**.
3. Tap **Version** (or **Software Information**, depending on your ColorOS version).
4. Find **Build Number**.
5. Tap **Build Number 7 times**.
6. Enter your lock screen PIN/password if prompted.
7. You should see a message like:

"You are now a developer!"

Step 2: Enable USB Debugging

1. Go back to **Settings**.
2. Open **Additional Settings**.
3. Tap **Developer Options**.
4. Turn on:
    - **Developer options**
    - **USB debugging**
5. When prompted, tap **OK**.

Step 3: Connect to Your PC

1. Connect your phone using a **USB data cable**.
2. On your phone, when the USB options appear, choose:
    - **File Transfer (MTP)** (not "Charge only").
3. A pop-up will appear:

Allow USB debugging?

4. Check:
    - **Always allow from this computer**
5. Tap **Allow**.

Step 4: Run Your App

1. Open **Android Studio**.
2. Wait a few seconds.
3. Your phone should appear in the device dropdown at the top (e.g., **OPPO Reno 10x Zoom**).
4. Select it and click the ▶ **Run** button.

OR

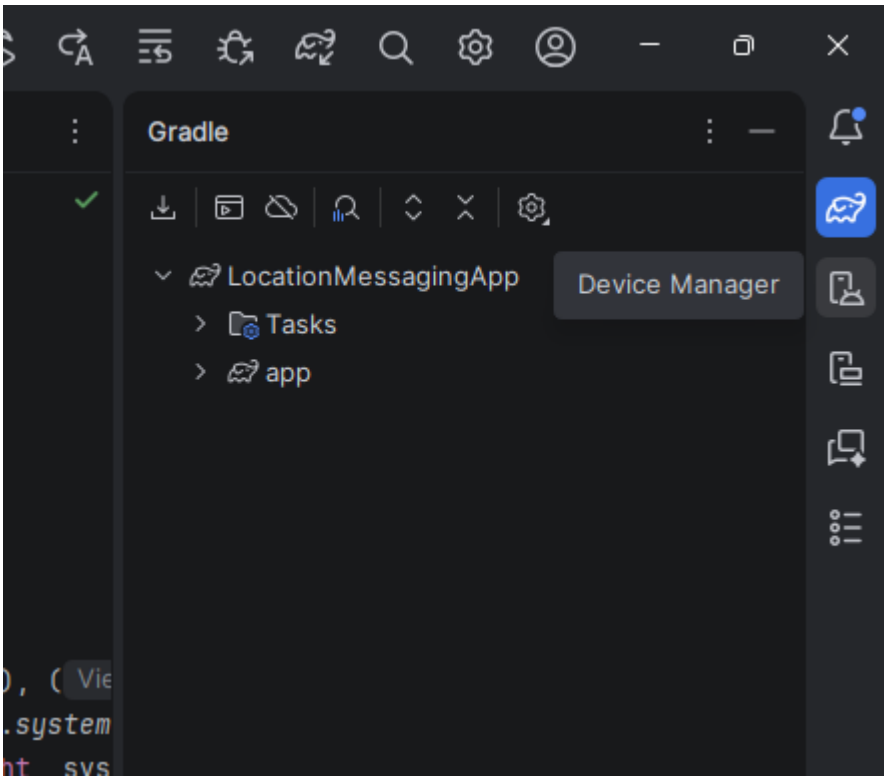
Create an Android Emulator (AVD)

This is required to run the app on your PC.

At the top, you should see an option like:

- **Device Manager**
- **Add Device**

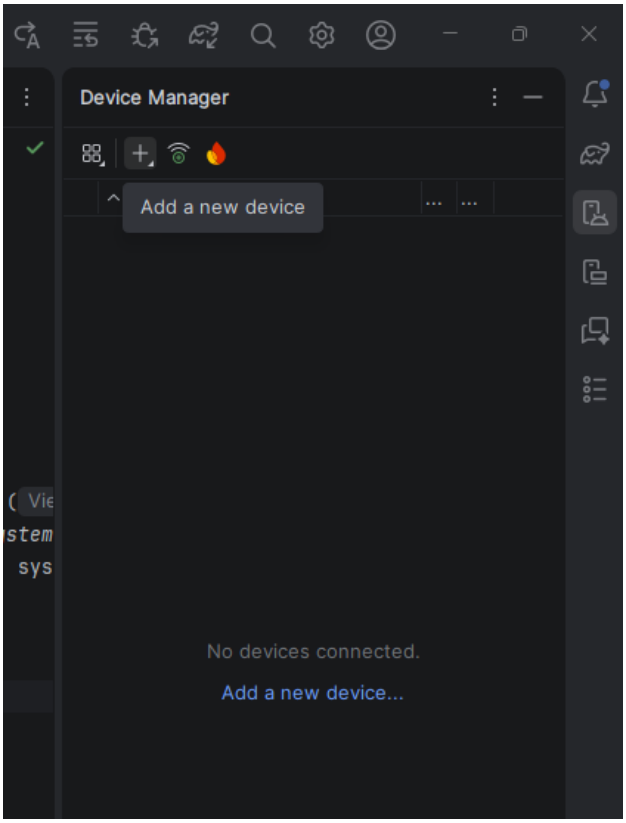
Click it.

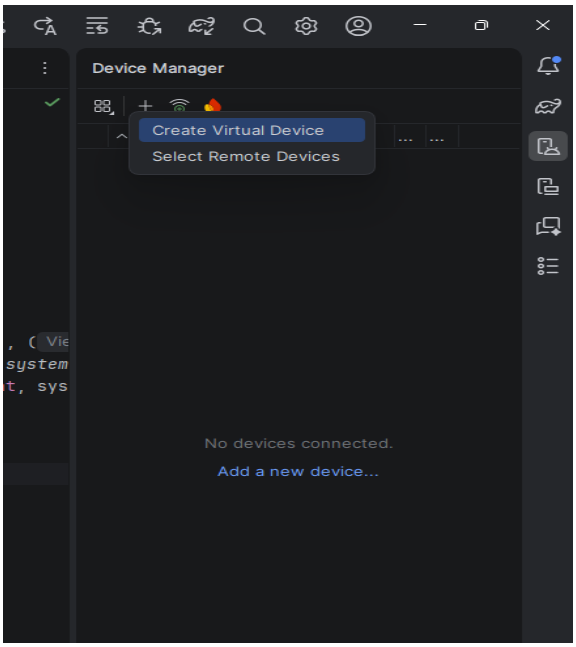


Click:

+ Add a new device

Now, You're in the Device Manager.



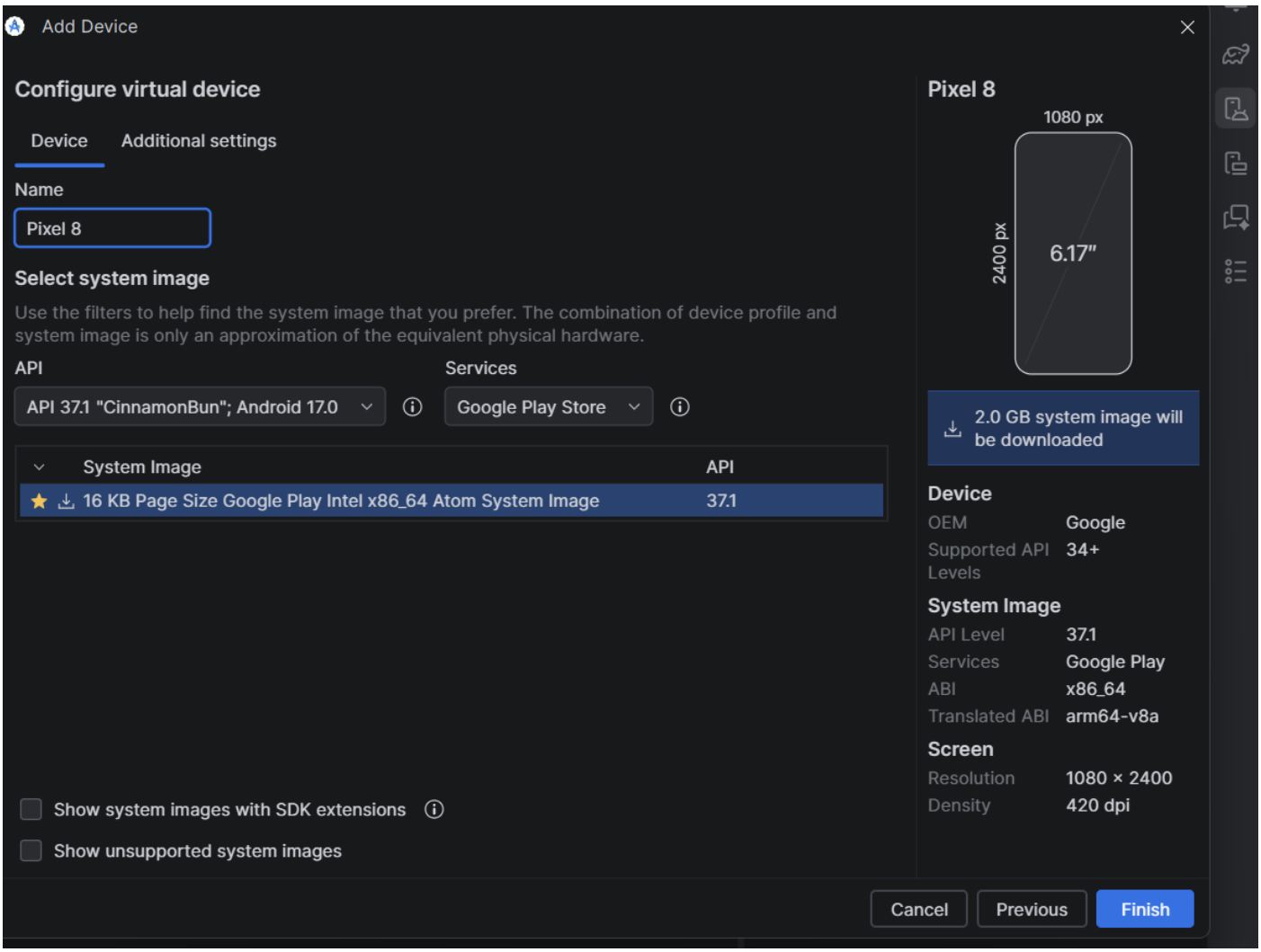


For this practical I recommend using **Pixel 8** because:

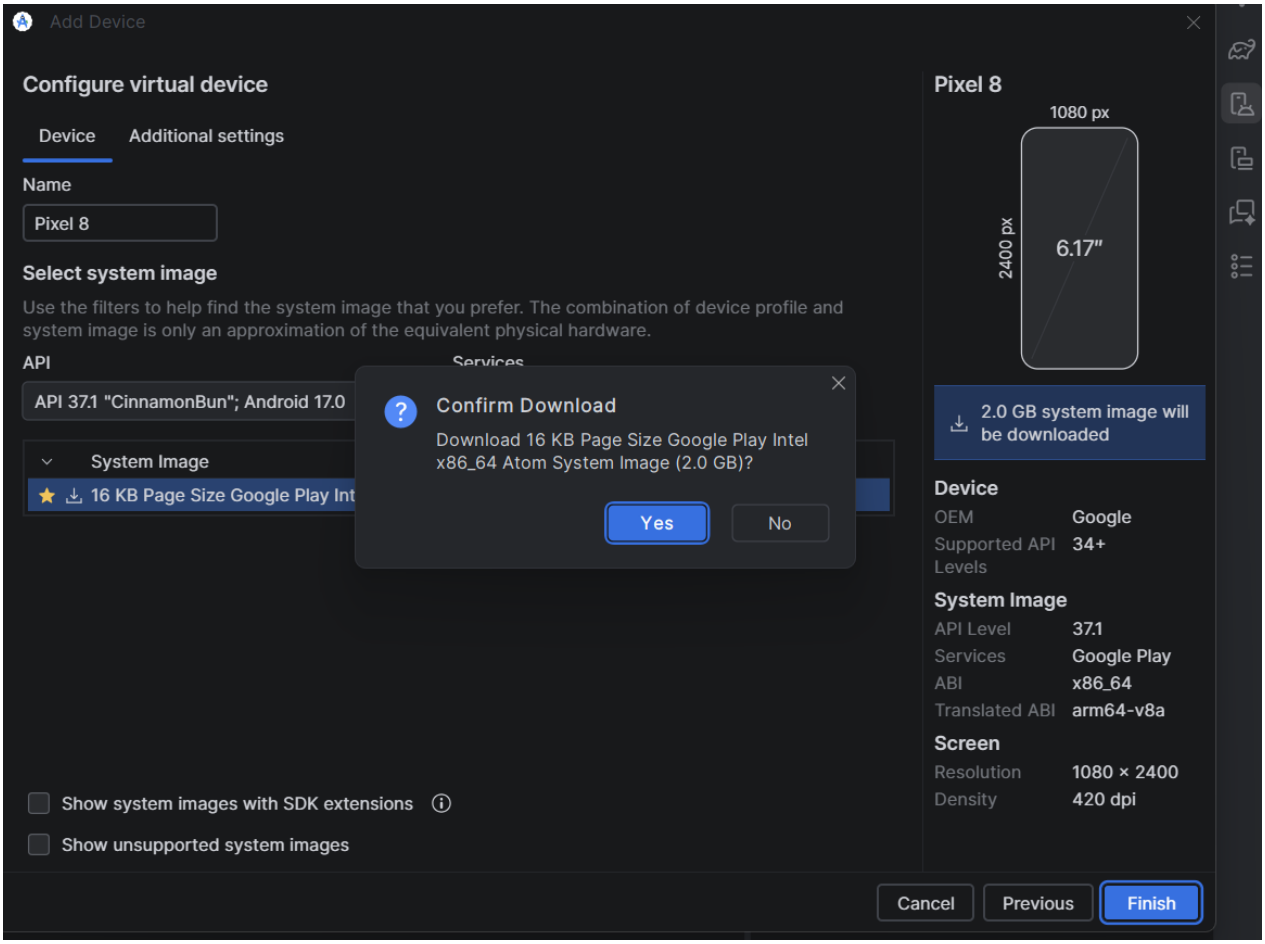
- It supports **API 34+**
- It's stable
- Google officially supports it
- Good for demonstrating GPS and location services
- Better than older Pixel 6 images in the latest Android Studio

Select: **Pixel 8**

Then click **Finish.**



Since you're using **Android Studio Quail 1**, Google has changed the interface. Instead of downloading first and then finishing, it downloads automatically after you click **Finish**.



Shows:

- Device: **Pixel 8**
- API: **37.1 (Android 17 - Cinnamon Bun)**
- Google Play System Image
- 2.0 GB will be downloaded

Click: **Yes**

Android Studio will start downloading the **2.0 GB system image**.

It may take **10–30 minutes**, depending on your internet speed.

Step 7:

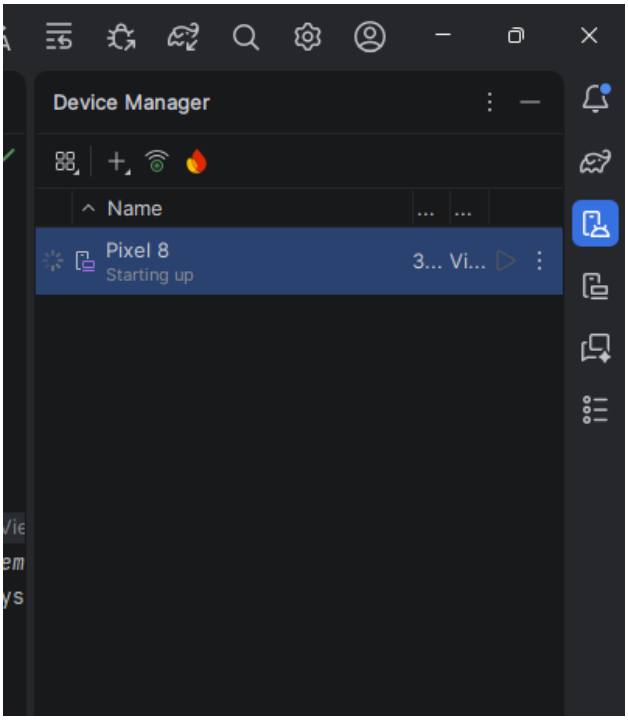
After the download completes, you'll see something like:

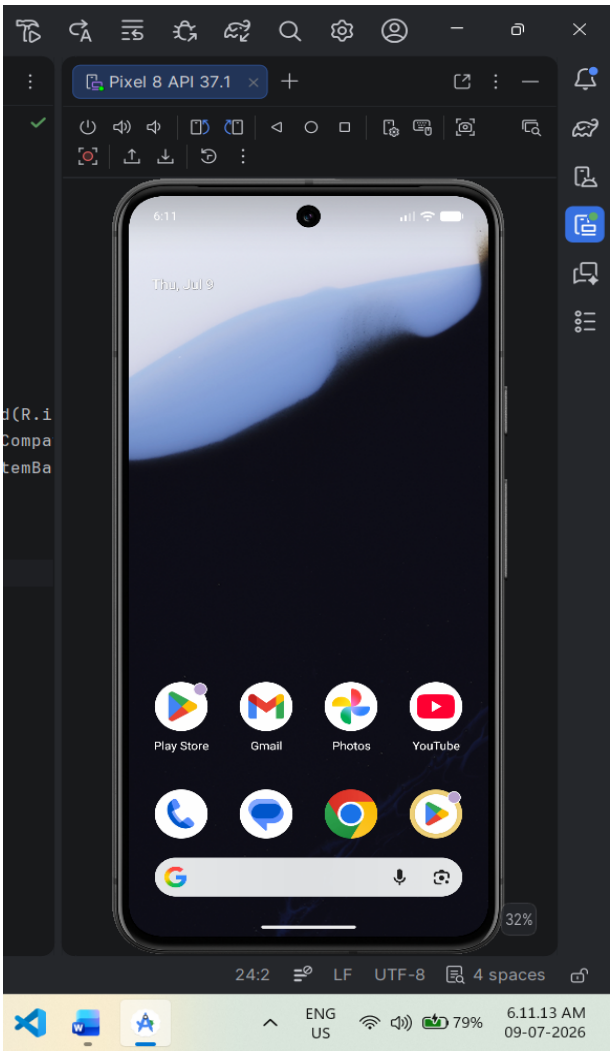
```
Device Manager

Pixel 8
▶
```

Click the ▶ **(Play)** button to start the emulator.

Asst. Professor Ms. Harshita Pandey





Step8: **MainActivity.java**

```
package com.example.locationmessagingapp;
```

```
import android.Manifest;
import android.content.pm.PackageManager;
import android.os.Bundle;
import android.widget.Button;
import android.widget.TextView;

import androidx.annotation.NonNull;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.app.ActivityCompat;

import com.google.android.gms.location.FusedLocationProviderClient;
import com.google.android.gms.location.LocationServices;
import com.google.android.gms.location.Priority;

public class MainActivity extends AppCompatActivity {

    private TextView txtLatitude, txtLongitude, txtMessage;

    private FusedLocationProviderClient fusedLocationClient;

    private static final int LOCATION_PERMISSION_REQUEST = 100;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        Button btnLocation = findViewById(R.id.btnLocation);
        txtLatitude = findViewById(R.id.txtLatitude);
        txtLongitude = findViewById(R.id.txtLongitude);
        txtMessage = findViewById(R.id.txtMessage);

        fusedLocationClient = LocationServices.getFusedLocationProviderClient(this);

        btnLocation.setOnClickListener(v -> getCurrentLocation());
    }
```

```
  private void getCurrentLocation() {

      if (ActivityCompat.checkSelfPermission(
              this,
              Manifest.permission.ACCESS_FINE_LOCATION)
              != PackageManager.PERMISSION_GRANTED) {

          ActivityCompat.requestPermissions(
                  this,
                  new String[]{
                          Manifest.permission.ACCESS_FINE_LOCATION,
                          Manifest.permission.ACCESS_COARSE_LOCATION
                  },
                  LOCATION_PERMISSION_REQUEST
          );
          return;
      }

      fusedLocationClient.getCurrentLocation(
              Priority.PRIORITY_HIGH_ACCURACY,
              null
      ).addOnSuccessListener(this, location -> {

          if (location != null) {

              double latitude = location.getLatitude();
              double longitude = location.getLongitude();

              txtLatitude.setText(getString(R.string.latitude, latitude));
              txtLongitude.setText(getString(R.string.longitude, longitude));

              if (latitude >= 18.8 && latitude <= 19.5 &&
                      longitude >= 72.7 && longitude <= 73.2) {

                  txtMessage.setText(R.string.welcome_message);

              } else {
```

```
                    txtMessage.setText(R.string.outside_message);
                }

            } else {

                txtMessage.setText(R.string.unable_location);
            }
        });
    }

    @Override
    public void onRequestPermissionsResult(
            int requestCode,
            @NonNull String[] permissions,
            @NonNull int[] grantResults) {

        super.onRequestPermissionsResult(requestCode, permissions, grantResults);

        if (requestCode == LOCATION_PERMISSION_REQUEST) {
            if (grantResults.length > 0 &&
                    grantResults[0] == PackageManager.PERMISSION_GRANTED) {

                getCurrentLocation();

            } else {

                txtMessage.setText(R.string.permission_denied);
            }
        }
    }
}
```

Step9: **activity_main.xml**

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
android:layout_width="match_parent" android:layout_height="match_parent"
android:orientation="vertical" android:gravity="center" android:padding="20dp">

   <TextView android:id="@+id/txtTitle" android:layout_width="wrap_content"
android:layout_height="wrap_content" android:text="Location Based Messaging App"
android:textSize="24sp" android:textStyle="bold" android:layout_marginBottom="30dp"/>

   <Button android:id="@+id/btnLocation" android:layout_width="wrap_content"
android:layout_height="wrap_content" android:text="Get Current Location"/>

   <TextView android:id="@+id/txtLatitude" android:layout_width="wrap_content"
android:layout_height="wrap_content" android:text="Latitude : " android:textSize="18sp"
android:layout_marginTop="25dp"/>

   <TextView android:id="@+id/txtLongitude" android:layout_width="wrap_content"
android:layout_height="wrap_content" android:text="Longitude : "
android:textSize="18sp"/>

   <TextView android:id="@+id/txtMessage" android:layout_width="wrap_content"
android:layout_height="wrap_content" android:text="Location Message"
android:textSize="20sp" android:textStyle="bold" android:layout_marginTop="25dp"/>

</LinearLayout>
```

Step10: **libs.versions.toml**

```
[versions]

agp = "9.2.1"

junit = "4.13.2"

junitVersion = "1.3.0"

espressoCore = "3.7.0"

appcompat = "1.7.1"

material = "1.14.0"

activityKtx = "1.13.0"

constraintlayout = "2.2.1"

playServicesLocation = "21.3.0"


[libraries]

junit = { group = "junit", name = "junit", version.ref = "junit" }

ext-junit = { group = "androidx.test.ext", name = "junit", version.ref = "junitVersion"
}

espresso-core = { group = "androidx.test.espresso", name = "espresso-core", version.ref
= "espressoCore" }

appcompat = { group = "androidx.appcompat", name = "appcompat", version.ref =
"appcompat" }

material = { group = "com.google.android.material", name = "material", version.ref =
"material" }

activity-ktx = { group = "androidx.activity", name = "activity-ktx", version.ref =
"activityKtx" }
```

```
constraintlayout = { group = "androidx.constraintlayout", name = "constraintlayout", version.ref = "constraintlayout" }

play-services-location = { group = "com.google.android.gms", name = "play-services-location", version.ref = "playServicesLocation" }


[plugins]

android-application = { id = "com.android.application", version.ref = "agp" }
```

Step 11: **build.gradle.kts (:app)**

```
plugins {
    alias(libs.plugins.android.application)
}


android {
    namespace = "com.example.locationmessagingapp"
    compileSdk {
        version = release(36) {
            minorApiLevel = 1
        }
    }

    defaultConfig {
        applicationId = "com.example.locationmessagingapp"
        minSdk = 24
        targetSdk = 36
        versionCode = 1
        versionName = "1.0"

        testInstrumentationRunner = "androidx.test.runner.AndroidJUnitRunner"
    }

    buildTypes {
        release {
            optimization {
                enable = false
            }
        }
```

```
    }
    compileOptions {
        sourceCompatibility = JavaVersion.VERSION_11
        targetCompatibility = JavaVersion.VERSION_11
    }
}

dependencies {
    implementation(libs.activity.ktx)
    implementation(libs.appcompat)
    implementation(libs.constraintlayout)
    implementation(libs.material)
    implementation("com.google.android.gms:play-services-location:21.3.0")
    testImplementation(libs.junit)
    androidTestImplementation(libs.espresso.core)
    androidTestImplementation(libs.ext.junit)
}
```

Step 12: **build.gradle.kts (LocationMessagingApp)**

```
// Top-level build file where you can add configuration options common to all sub-
projects/modules.
plugins {
    alias(libs.plugins.android.application) apply false
}
```

Step 13: **AndroidManifest.xml**

```
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:tools="http://schemas.android.com/tools">
    <uses-permission android:name="android.permission.ACCESS_FINE_LOCATION"/>
    <uses-permission android:name="android.permission.ACCESS_COARSE_LOCATION"/>
    <application android:allowBackup="true"
android:dataExtractionRules="@xml/data_extraction_rules"
android:fullBackupContent="@xml/backup_rules" android:icon="@mipmap/ic_launcher"
android:label="@string/app_name" android:roundIcon="@mipmap/ic_launcher_round"
android:supportsRtl="true" android:theme="@style/Theme.LocationMessagingApp">
        <activity android:name=".MainActivity" android:exported="true"
android:windowSoftInputMode="adjustResize">
            <intent-filter>
```

```
                <action android:name="android.intent.action.MAIN"/>
                <category android:name="android.intent.category.LAUNCHER"/>
            </intent-filter>
        </activity>
    </application>
</manifest>
```

Step 13.1: **strings.xml**

Replace your app > res > values > strings.xml with:

```
<resources>

    <string name="app_name">LocationMessagingApp</string>

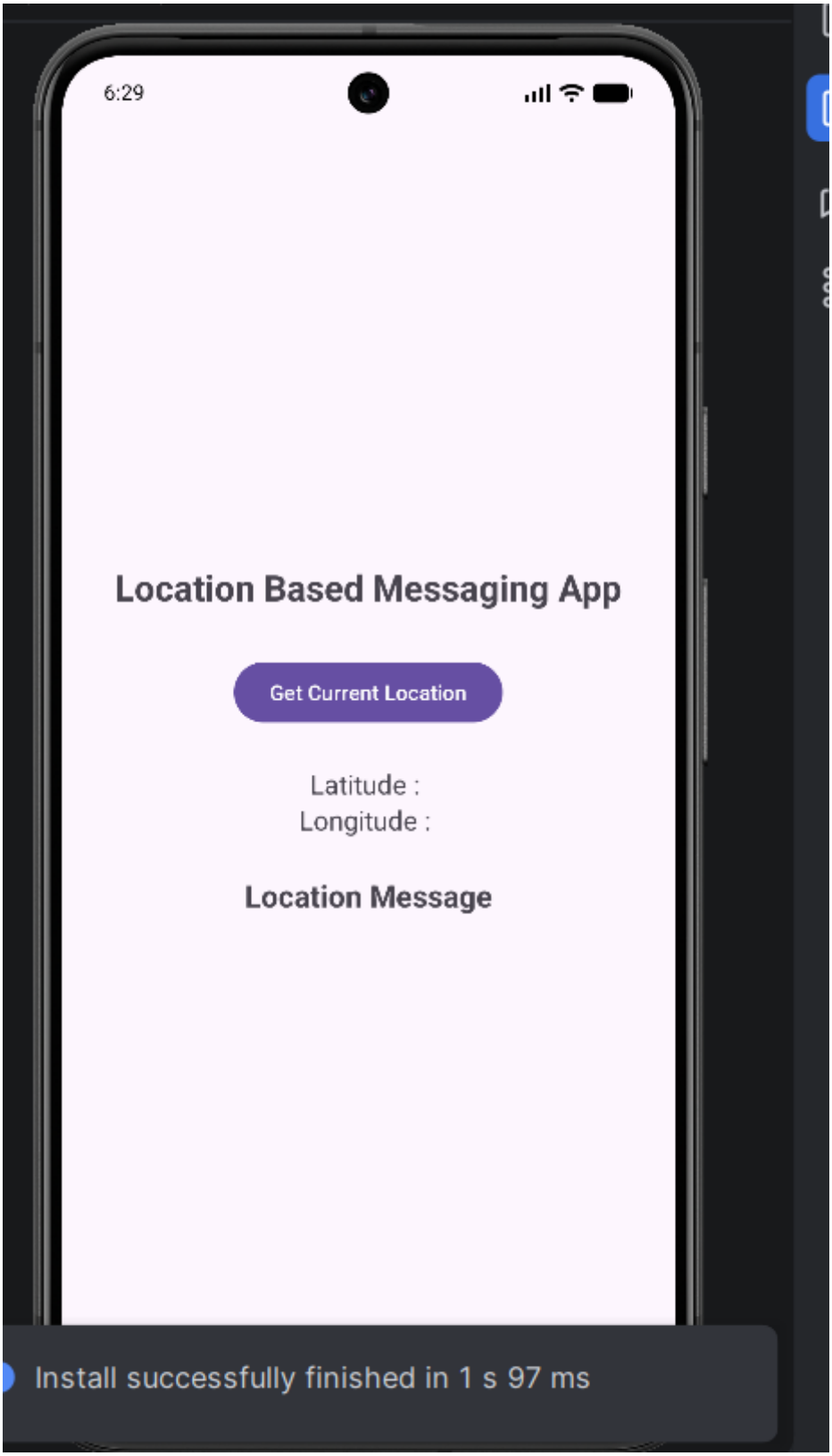
    <string name="latitude">Latitude : %1$.6f</string>
    <string name="longitude">Longitude : %1$.6f</string>

    <string name="welcome_message">Welcome to Mumbai Harshita</string>
    <string name="outside_message">You are outside Mumbai</string>
    <string name="unable_location">Unable to fetch location.</string>
    <string name="permission_denied">Location Permission Denied</string>

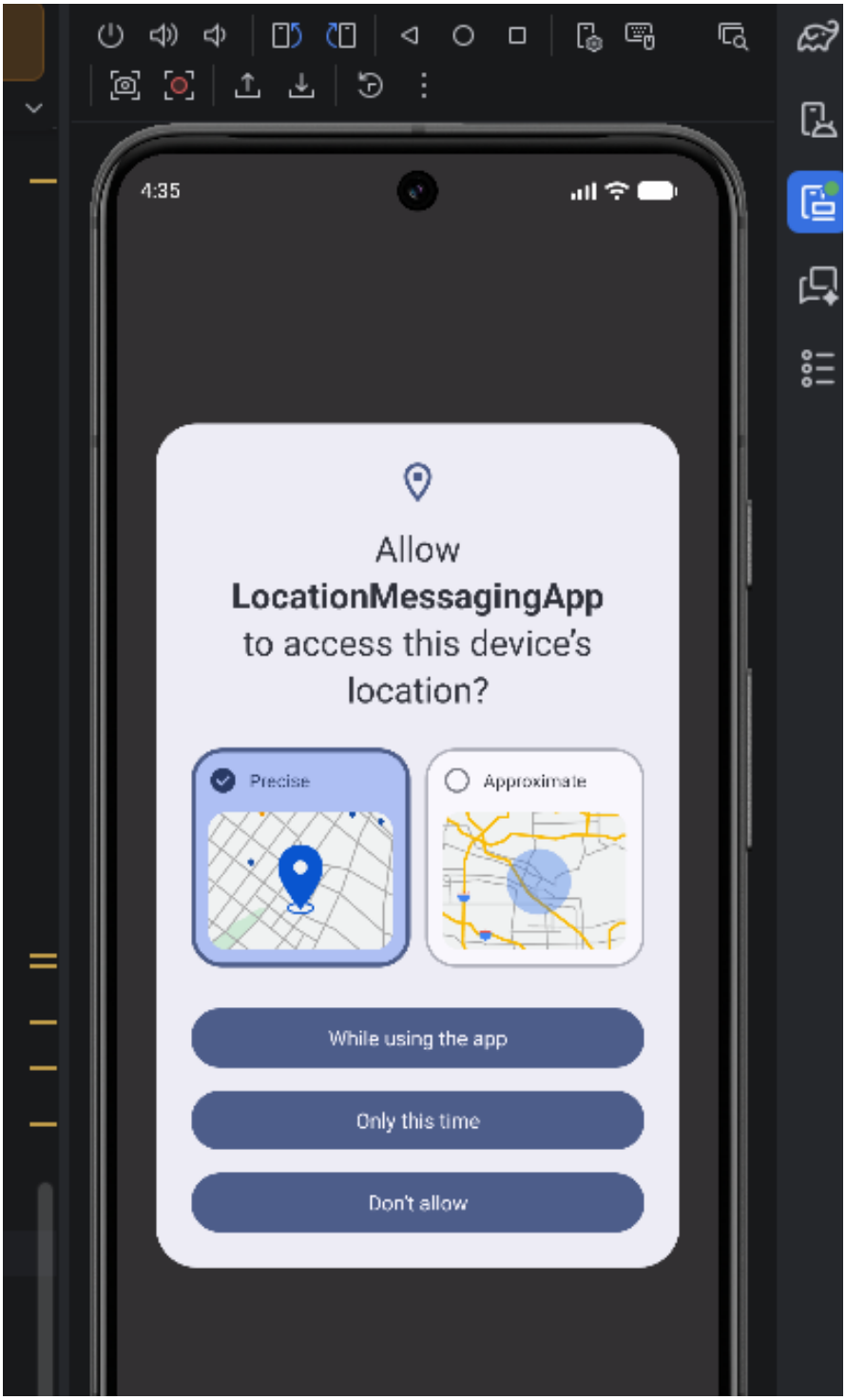
</resources>
```

Step 14:

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